

Reflective Lantern - Innovation Report

New Project Built

Name: Quake-Net

Branch: <https://github.com/atharvadevne123/reflective-lantern/tree/innovation/quake-net>

Draft PR: <https://github.com/atharvadevne123/reflective-lantern/pull/51>

Date: 2026-08-21

Inspired by: Portfolio gap analysis - seismology was the largest untouched ML domain across 21 prior projects (HN API and GitHub trending unreachable: egress proxy returns 403 on non-proxied HTTP)

What it does: Seismic event magnitude prediction and aftershock forecasting API using an XGBoost + RandomForest voting ensemble, modified Omori law decay modelling, KS-test and PSI drift monitoring, IsolationForest anomaly detection, FAISS similarity search and champion/challenger automated retraining.

Mandatory Tech Stack Coverage: 100% (10/10)

- [X] Python 3.11
- [X] FastAPI (11 endpoints)
- [X] XGBoost + RandomForest ensemble
- [X] sklearn Pipeline (6 stages, 47 features)
- [X] Model monitoring (KS + PSI drift)
- [X] Docker + docker-compose + PostgreSQL
- [X] SQLAlchemy (SQLite dev / Postgres prod)
- [X] pytest suite (285 tests, 11 modules)
- [X] .env.example
- [X] GitHub Actions CI (ruff + pytest)

Optional Enhancements: 7 of 4 required

- [X] FAISS similarity search (brute-force fallback)
- [X] Airflow champion/challenger retraining DAG
- [X] Time-series forecasting (modified Omori law)
- [X] Anomaly detection (IsolationForest + z-score + IQR)
- [X] Automated retraining with R2 gate
- [X] Drift detection (KS-test + PSI)
- [X] 5-fold cross-validation

Model Quality

CV R2 (5-fold): 0.8216 +/- 0.0211

Held-out R2: 0.8029

RMSE: 0.2642 magnitude units

MAE: 0.2080 magnitude units

Engineered features: 47 (from 8 raw columns)

Note: no proprietary seismic catalogue ships with the repo. Training data is synthesised from a Richter-like relationship with Gaussian noise, so these metrics describe recovery of a known physical relationship, not accuracy on real catalogue data.

Bugs Found and Fixed During the Build

1. DropCategoricalColumns selected by dtype == object, missing string columns under the PyArrow-backed pandas default; non-numeric data would have reached StandardScaler.
2. Champion/challenger gate read the champion R2 from metrics.json AFTER training overwrote it, comparing the challenger against itself - every model promoted, including regressions.
3. Ad-hoc similarity index builds overwrote the persisted FAISS index the API serves from; persistence is now opt-in.
4. n_features reported the raw input width (8) instead of the post-pipeline width (47).
5. model_metrics table was defined but never written to - training history lost on every retrain.

Project Stats

Files created: 50

Tests written: 285 (all passing)

Commits: 62

Lint: ruff check + ruff format both clean

Publishing Note

Standalone repo creation returned 403: this session's GitHub access is bound to reflective-lantern. The project ships on branch `innovation/quake-net` with draft PR #51. A one-click publisher workflow (`.github/workflows/bootstrap-quake-net.yml`) can be run from the Actions tab with a repo-scope PAT to create the Quake-Net repo, push all 60 project commits as main via subtree split, tag v1.0.0, create the release, and upload the wheel.